

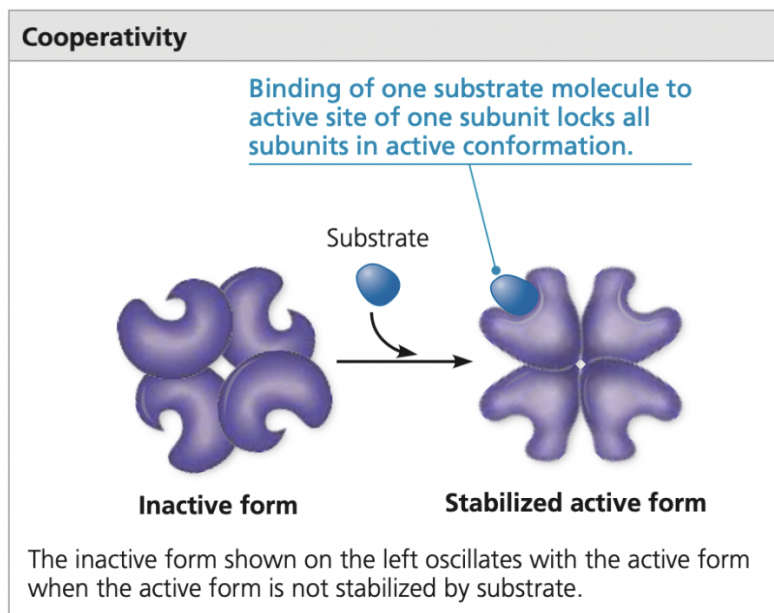


Figure 4.5 A diagram depicting positive cooperativity in an enzyme. (Source: Campbell Biology)

4.3.4 Feedback Inhibition

One interesting phenomenon found in many cellular processes involving enzymes is **feedback inhibition**. The products of an enzyme-controlled reaction eventually go back and inhibit its creation.

One example of feedback inhibition can be found in the process of cellular respiration. **Pyruvate dehydrogenase** is an enzyme which catalyzes the formation of **Acetyl-CoA** in the cell. The molecule **Acetyl-CoA** then inhibits **pyruvate dehydrogenase**. This may seem counter-intuitive at first, however it leads to overall regulation of Acetyl-CoA levels; when too much of the molecule is present, the enzyme involved in its creation is inhibited, and when too little of the molecule is present, the enzyme is uninhibited and free to catalyze more reactions.

Feedback inhibition can also reach back many steps in a chemical pathway. **ATP**, the source of energy for the cell and one of the final products of cellular respiration, is able to inhibit an enzyme involved in the very beginning steps of the entire process.

All of these types of enzyme control play a part in maintaining the homeostasis of a cell. Through allosteric regulation, competitive and noncompetitive inhibitors and activators, feedback inhibition and activation, cooperativity, and more, the multitudes of processes in a cell are catalyzed and regulated.

4.4 Enzyme Kinetics

Enzyme kinetics is the study of the rate of catalyzed reactions, and how different factors can affect the impact an enzyme has on a particular chemical reaction.

4.4.1 Michaelis-Menten Kinetics

The Michaelis-Menten equation is an important tool used by biologists to determine how quickly an enzyme-catalyzed reaction will proceed, and is as follows:

$$v = \frac{V_{max}a}{K_M + a}$$

where v represents the velocity of the reaction, V_{max} represents the maximum rate of the reaction, a represents the concentration of the substrate, and K_M represents the Michaelis constant.

An enzyme catalyzes a reaction very quickly—substrates enter the active site, a reaction is catalyzed, and a product leaves the enzyme—up to millions of times a second. Most of the time, an enzyme is idly waiting for a substrate to bind to its active site. By increasing the concentration of a substrate a in an environment, this wait time is decreased, and the enzyme can thus catalyze reactions quicker. At some point, increasing a will have no effect on the rate of a reaction, as the enzyme will have no time between individual catalyses. This is V_{max} . K_M is a constant which measures the affinity between the enzyme and its substrate. It is the substrate concentration at which the reaction rate is half of V_{max} .

It is important to note that there are other ways of increasing reaction rate than increasing a . Simply increasing the amount of enzymes will lead to a greater total reaction velocity. This is reflected in the Michaelis-Menten equation, where since enzyme concentration is increased, V_{max} is increased. Enzymes also have optimal temperature and pH levels where they work best, also reflected in the Michaelis-Menten equation by increasing V_{max} as conditions become closer to ideal.

The Michaelis-Menten equation for a chemical reaction can be graphed, creating a **Michaelis-Menten curve**, allowing biologists to visualize the change in v against a .

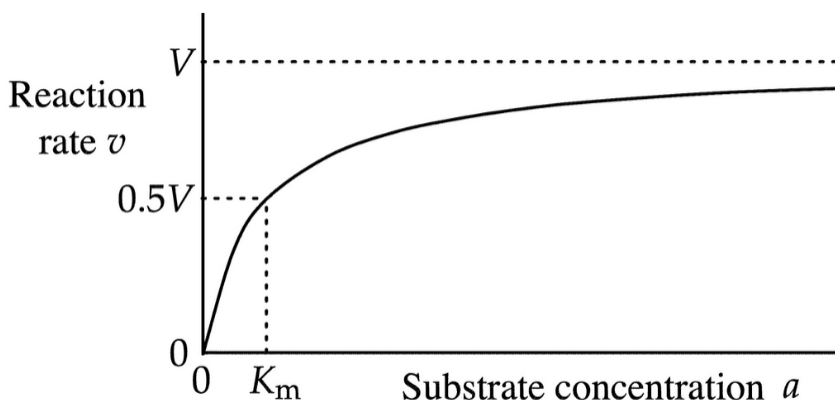
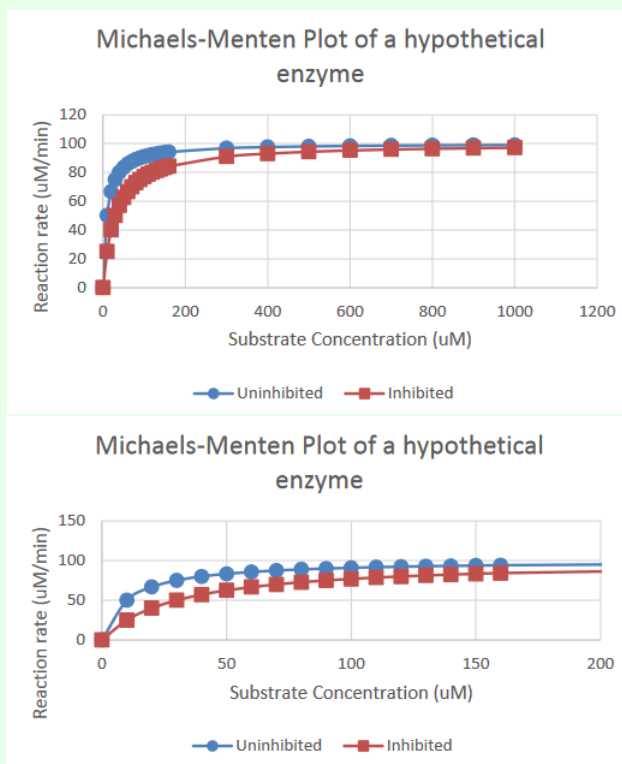


Figure 4.6 A Michaelis-Menten curve. (Source: Athel Cb)

Example 4.2 (USABO Semifinal Exam 2015) Hypothetical data for an enzyme on its reaction kinetics were collected in the presence and absence of an inhibitor. Assuming that the enzyme is properly modeled using Michaelis-Menten kinetics, Use the two plots below

to answer the following questions.



Note that the plots show the same data, however, the horizontal axes are scaled differently.

What is the maximum reaction rate possible for the uninhibited enzyme under these conditions?

- (A) 10 uM/min
- (B) 40 uM/min
- (C) 80 uM/min
- (D) 100 uM/min
- (E) 120 uM/min

Solution: In this question, we are being asked to find the V_{max} of the uninhibited enzyme. We can thus ignore the red line representing the kinetics of the inhibited enzyme, and focus solely on the blue one. Recall that the Michaelis-Menten equation is:

$$v = \frac{V_{max}a}{K_M + a}$$

In the case of this graph, only substrate concentration is being increased. As a approaches infinity, v eventually approaches V_{max} , forming a horizontal asymptote at $v = V_{max}$. When looking at the graph, there is a clear horizontal asymptote at $v = 100$ uM/min. The answer is thus D.

4.4.2 Lineweaver-Burk Plot

The **Lineweaver-Burk plot**, also known as the **double reciprocal plot**, is a transformation of the Michaelis-Menten equation. From its name, the reciprocals of both sides of the equation are taken and plotted. The transformation equation is as follows:

$$\frac{1}{v} = \frac{K_M}{V_{max}} \cdot \frac{1}{a} + \frac{1}{V_{max}}$$

Note its resemblance to the general linear equation $y = mx + b$. When plotting a Lineweaver-Burk plot with $1/V_{max}$ over $1/a$, the result is a linear output with a slope of K_M/V_{max} and y-intercept of $1/V_{max}$.

Lineweaver-Burk plots are used especially commonly to show the effects of enzyme inhibition.

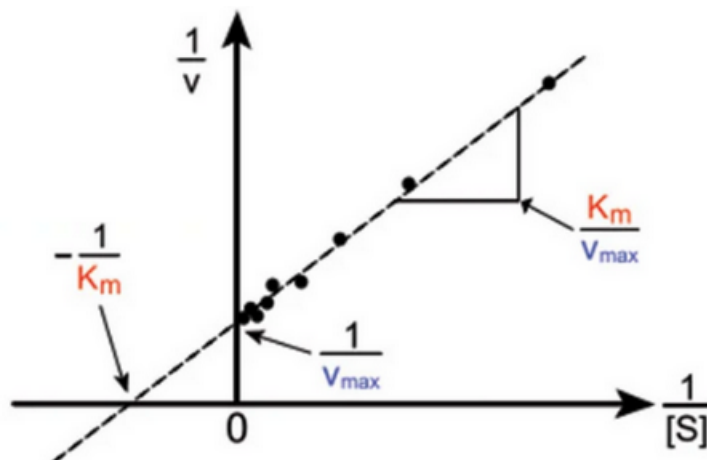


Figure 4.7 A Lineweaver-Burk plot. Note that substrate concentration is represented by $[S]$ in this depiction. (Source: PhD Nest)

Example 4.3 (IMDO Exam 2022) In the Lineweaver Burk plot, what is the x-intercept?

- (A) V_{max}
- (B) K_M
- (C) $1/V_{max}$
- (D) $-1/K_M$

Solution: Recall the transformed Michaelis-Menten equation used for a Lineweaver-Burk double reciprocal plot:

$$\frac{1}{v} = \frac{K_M}{V_{max}} \cdot \frac{1}{a} + \frac{1}{V_{max}}$$

Substituting y for $\frac{1}{v}$ and x for $\frac{1}{a}$ leaves us with the new equation:

$$y = \frac{K_M}{V_{max}}x + \frac{1}{V_{max}}$$

This equation is similar in form to the general straight line equation $y = mx + b$. The

x-intercept of a graph is the point at which $y = 0$, so setting $y = 0$ and solving for x will lead to the x-intercept.

$$0 = \frac{K_M}{V_{max}}x + \frac{1}{V_{max}}$$

$$\frac{K_M}{V_{max}} = -\frac{1}{V_{max}}$$

$$x = -\frac{1}{K_M}$$

The x-intercept of a Lineweaver-Burk plot is $-\frac{1}{K_M}$, and thus the answer is D.

4.4.3 Effects of Inhibition

Inhibition affects the K_M and V_{max} of an enzyme-catalyzed reaction, which can then be portrayed effectively using Michaelis-Menten curves or Lineweaver-Burk plots.

Competitive inhibition leads to **increased K_M** and has **no effect on V_{max}** .

- In competitive inhibition, an inhibitor molecule binds to the active site of an enzyme, effectively reducing the total amount of active enzymes available for binding by the substrate. A higher substrate concentration is required to achieve half of the maximum reaction rate, so K_M is increased.
- Since the competitive inhibitor does not affect the catalytic properties of an enzyme, V_{max} is unchanged.

Noncompetitive inhibition has **no effect on K_M** and leads to **decreased V_{max}** .

- In noncompetitive inhibition, an inhibitor molecule binds to the allosteric site of an enzyme, reducing its catalytic efficiency. Since it does not affect the affinity of the enzyme for its substrate, K_M is unchanged.
- Since the noncompetitive inhibitor affects the catalytic properties of an enzyme, V_{max} is decreased.

It is also important to note the existence of **uncompetitive inhibition**, which leads to both **decreased K_M** and **decreased V_{max}** .

- In uncompetitive inhibition, the inhibitor molecule is only able to bind to enzyme-substrate complexes. When bound, it increases the tightness of the grip between the enzyme and the substrate, essentially trapping the substrate to the enzyme. This leads to a decreased K_M , increasing the affinity of the substrate to the enzyme.
- When inhibiting an enzyme-substrate complex, the substrate is trapped and cannot release as easily. This decreases the catalytic efficiency of the enzyme, and thus V_{max} is decreased.

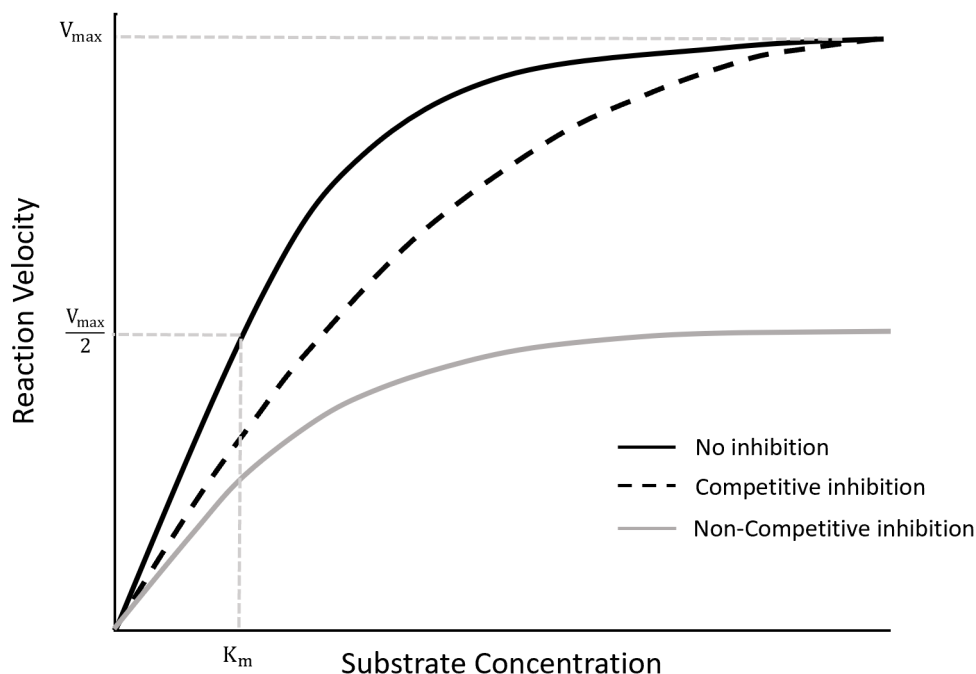


Figure 4.8 A Michaelis-Menten curve of an enzyme-catalyzed reaction under normal conditions, competitive inhibition, and noncompetitive inhibition. (Source: Hansehan)

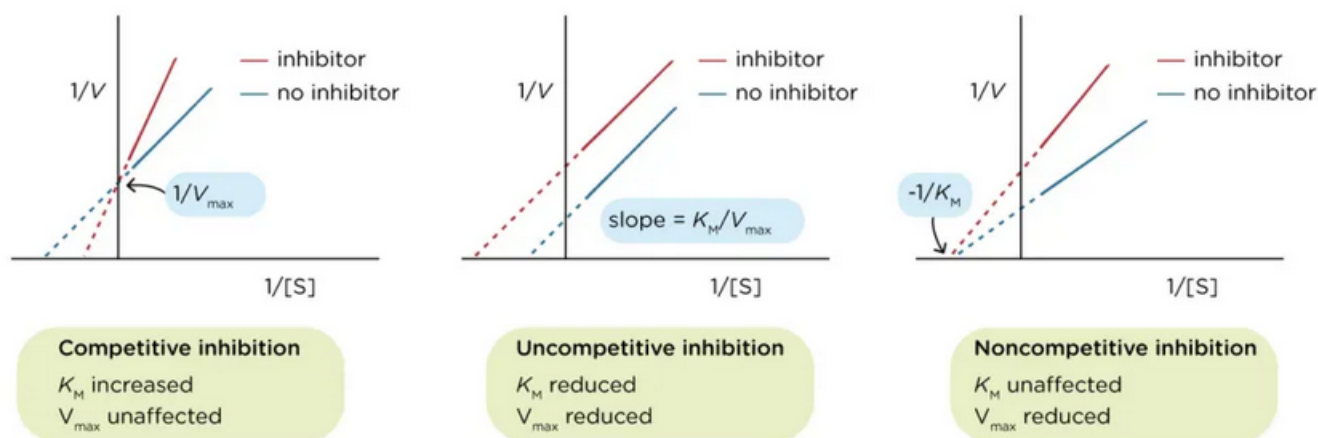


Figure 4.9 Multiple Lineweaver-Burk plots depicting the effects of competitive, uncompetitive, and noncompetitive inhibition on an enzyme-catalyzed reaction. (Source: Jack Westin)

Example 4.4 (Adapted from IMDO Exam 2022) Which of the following is found in non-competitive inhibition of an enzyme?

- (A) Increased V_{max}
- (B) Unaffected V_{max}
- (C) Increased K_M
- (D) Unaffected K_M

Solution: It may be helpful to think about how each type of inhibition affects an enzyme and deduce its effects on K_M and V_{max} from there. For example, remember that noncompetitive inhibition is a form of allosteric regulation where the inhibitor molecule does not bind to the active site of an enzyme. Because of this, noncompetitive inhibition has no effects on K_M , as it doesn't affect the affinity of the enzyme to its substrate. The answer is thus D.

5 Conclusion

5.1 Closing Statement

The concepts which have been outlined in this paper have hopefully been enlightening to a new reader who has not yet learned the laws of thermodynamics and the universe, and opened their eyes to the depth of biology and how far life has come. As we study physics and make observations of the world around us, we realize the pure, unbreakable laws which govern our lives, of entropy and energy. As we study biology, we see the miraculous ways our cells have evolved in order to overcome and bypass such laws. We exist in a world constantly breaking apart, and eventually the entropic heat death of the universe will occur, but our cells and life as we know it have seemed to temporarily overcome entropy and have built our house of cards.